

Demonstration of Proposed Features

Date	4 July 2026
Team ID	
Project Name	Voice Based Communication Understanding Analyzer
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document presents the proposed features of the **Voice-Based Concept Understanding Analyser (VBCUA)** and their implementation status. It verifies that the planned functionalities were successfully developed, tested, and demonstrated during the project. The implemented features collectively provide an AI-powered solution for evaluating spoken conceptual understanding through speech recognition, semantic analysis, audio processing, intelligent scoring, and automated reporting.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Audio Upload	Allows users to upload WAV/MP3 audio files for analysis.	Implemented	Yes	Successfully accepts audio input.
2	Speech-to-Text Transcription	Converts uploaded speech into text using OpenAI Whisper.	Implemented	Yes	Accurate transcription demonstrated.
3	Semantic Similarity Analysis	Compares the spoken explanation with reference content using Sentence-BERT.	Implemented	Yes	Displays semantic similarity score.
4	Audio Feature Extraction	Extracts speech features such as pauses, RMS energy, duration, and fluency using Librosa.	Implemented	Yes	Speech quality metrics displayed successfully.
5	Intelligent Scoring & Feedback	Generates conceptual understanding score and qualitative feedback (Strong, Moderate, Poor).	Implemented	Yes	AI-generated evaluation demonstrated.
6	PDF Report Generation	Generates a downloadable report containing transcript, scores, waveform, and feedback.	Implemented	Yes	PDF report successfully generated and downloaded.

Feature Implementation Summary:

Metric	Count / Value
Total Features Proposed	6

Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%